ECE 310 Fall 2026
Lecture 4: Convolution and impulse response

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Learning Objectives

- Define the impulse response of a discrete-time system.
- Show how the response of LTI systems to input signals can be fully described by their impulse response.
- Define the convolution between two discrete-time signals and key properties of convolution.
- Compute the convolution between two discrete-time signals by hand.



Impulse Response

Impulse response

The *impulse response* $h[n]$ of a discrete-time system T is the response or output of the system to a unit impulse or Kronecker delta $\delta[n]$ input:

$$h[n] = T(\delta[n])$$

Let's consider a couple examples of impulse responses.

- $y_1[n] = x[n] - x[n-1]$

$x[n] = \delta[n] \rightarrow \boxed{T} \rightarrow y_1[n] = h_1[n]$

$y[-1] = x[-1] - x[-2] = \delta[-1] - \delta[-2] = 0$

$y[0] = x[0] - x[-1] = \delta[0] - \delta[-1] = 1$

$y[1] = x[1] - x[0] = \delta[1] - \delta[0] = -1$

$y[2] = x[2] - x[1] = \delta[2] - \delta[1] = 0$

$\rightarrow y_1[n] = h[n] = \{1, -1, 0, 0, \ldots\} = \delta[n] - \delta[n-1]$

- $y_2[n] = \text{median}\{x[n], x[n-1], x[n-2]\}$

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$$h[n] = T(\delta[n])$$

Let's consider a couple examples of impulse responses. $\delta[n] = \{\cdots, 0, 0, \underset{\uparrow}{1}, 0, 0, \cdots\}$

- $y_1[n] = x[n] - x[n-1]$
- $y_2[n] = \text{median}\{x[n], x[n-1], x[n-2]\}$

$x[n] = \delta[n] \rightarrow \boxed{T} \rightarrow y_2[n] = h_2[n]$

$y_2[-1] = \text{median}\{x[-1], x[-2], x[-3]\} = 0$

$y_2[0] = \text{median}\{x[0], x[-1], x[-2]\}$

$= \text{median}\{1, 0, 0\} = 0$

$y_2[1] = \text{median}\{0, 1, 0\} = 0$

$\Rightarrow y_2[n] = h_2[n] = 0$ for all n.

Signals as Combination of Impulses

- Consider $x[n] = \cos\left(\frac{\pi}{4}n\right)(u[n] - u[n-8]) = \left\{\underset{\uparrow}{1}, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right\}$.
  How can we write this signal as a linear combination of unit impulses?

Signals as Combination of Impulses

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  How can we write this signal as a linear combination of unit impulses?
- Each sample in $x[n]$ is associated with a location k and value $x[k]$. Thus,

$$x[n] = \delta[n] + \frac{\sqrt{2}}{2}\delta[n-1] - \frac{\sqrt{2}}{2}\delta[n-3] - \delta[n-4] - \frac{\sqrt{2}}{2}\delta[n-5] + \frac{\sqrt{2}}{2}\delta[n-7]$$



Signals as Combination of Impulses

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- In general, we may write any signal $x[n]$ as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

$x[4] = \sum_{k=-\infty}^{\infty} x[k]\delta[4-k]$

$= x[4]\delta[0]$ (1; zero for all other k)

$= x[4]$

System Response of Discrete-time Systems

Suppose we have a discrete-time system T with impulse response $h[n]$. How can we find the response of the system to arbitrary input signals?

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$\delta[n] \longrightarrow \boxed{T} \longrightarrow h[n]$

$a\delta[n] \longrightarrow \boxed{T} \longrightarrow$ (if satisfies homogeneity) $\rightarrow a h[n]$

$\delta[n-k] \longrightarrow \boxed{T} \longrightarrow$ (if system is time-invariant) $\rightarrow h[n-k]$

$\delta[n]+\delta[n-k] \longrightarrow \boxed{T} \longrightarrow$ (if system is TI and additive) $\rightarrow h[n]+h[n-k]$

$\sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \longrightarrow \boxed{T} \longrightarrow$ (if system is LTI) $\rightarrow \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

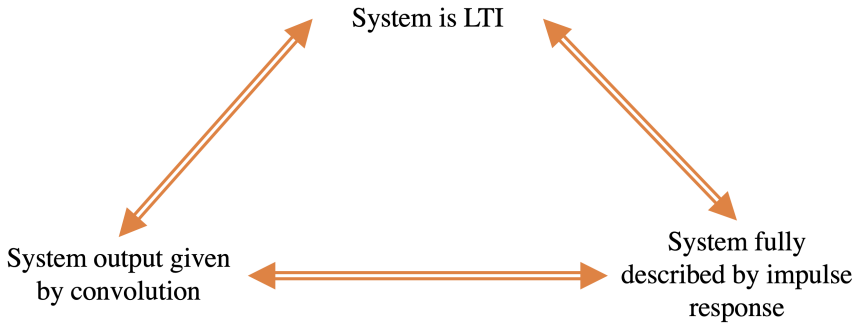
Linear + Time-invariant

Convolution

The system response $y[n]$ of an LTI system with impulse response $h[n]$ to input signal $x[n]$ is given by the *convolution* of $x[n]$ and $h[n]$, $y[n] = x[n] * h[n]$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k-\infty}^{\infty} h[k]x[n-k] \tag{1}$$





Key Properties of Convolution and Impulse Response

- **Commutativity**:

$$x[n] * h_1[n] * h_2[n] = x[n] * h_2[n] * h_1[n] = h_1 * x[n] * h_2[n] = \cdots$$

- **Distributivity**:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

- **Length**: The convolution of length-N $x[n]$ with length-M $h[n]$ results in the length-$(N + M - 1)$ signal $y[n] = x[n] * h[n]$.

- **Start Point and End Points**: Suppose $x[n]$ has start and end indices (first and last non-zero value locations) n_0 and n_1, respectively; $h[n]$ has start and end indices k_0 and k_1; then, $y[n] = x[n] * h[n]$ will have start and end indices:

$$\begin{aligned} \text{Start Index} &= n_0 + k_0 \\ \text{End Index} &= n_1 + k_1 \end{aligned}$$



Key Properties of Convolution and Impulse Response

- **Identity**:

$$x[n] * \delta[n] = x[n]$$

- **Causality**: An LTI system with impulse response $h[n]$ is causal if and only if

$$h[n] = 0, \ n < 0.$$

- **Stability**: An LTI system with impulse response $h[n]$ is BIBO stable if and only if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Computing Convolution

The general steps to *convolve* two sequences:

1. Flip one of the signals, e.g. $h[-k]$.
2. Shift the flipped signal by n samples to the right, e.g. $h[-(k-n)] = h[n-k]$,
3. Multiply the unchanged signal with the flipped-and-shifted signal element-wise, e.g. $z_n[k] = x[k]h[n-k]$.
4. Sum up the product of the two signal to obtain the output at index n, e.g. $y[n] = \sum_{k=-\infty}^{\infty} z_n[k]$.



Computing Convolution: Shift-and-Overlap

Suppose we would like to convolve the following two sequences:

$$x[n] = \{\underset{\uparrow}{1}, 2, 0, -4, -1\}, \quad h[n] = \{-2, \underset{\uparrow}{0}, 1\}$$

$n=0$ $n=0$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

- $x[n]$ is length-5
- $h[n]$ is length-3
- $g[n]$ is length-$(5+3-1)$ = length-7

$y[n] = \{-2, \underset{\uparrow}{-4}, 1, 10, 2, -4, -1\}$

	k	-2	-1	0	1	2	3	4	
	$x[k]$			1	2	0	-4	-1	$y[n]$
$n=-1$	$h[-(k+1)] = h[-1-k]$	1	0	-2					-2
$n=0$	$h[-k]$		1	0	-2				-4 ← $n=0$
$n=1$	$h[-(k-1)] = h[1-k]$			1	0	-2			1
$n=2$	$h[-(k-2)] = h[2-k]$				1	0	-2		10
	$h[3-k]$					1	0	-2	2
	$h[4-k]$						1	0	-4
	$h[5-k]$							1	-1

Computing Convolution: Matrix Method

Suppose we would like to convolve the following two sequences:

$$x[n] = \{\underset{\uparrow}{1}, 2, 0, -4, -1\}, \; h[n] = \{-2, \underset{\uparrow}{0}, 1\}$$

$x[n]$ starts @ $n=0$

$h[n]$ starts @ $n=-1$

$\downarrow$

$y[n]$ starts @ $n = 0+-1 = -1$

$y = Hx$

$x \in \mathbb{R}^5 \rightarrow$ length-5 vector

$y \in \mathbb{R}^7 \rightarrow$ length-$(5+3-1)$ = length-7

$H \in \mathbb{R}^{7\times 5}$

$y = Hx$

$$= \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ -4 \\ -1 \end{bmatrix} = \{-2, -4, 1, 10, 2, -4, -1\}$$

H $\quad$ x $\quad$ $n=0$

Computing Convolution: Matrix Method

Suppose we would like to convolve the following two sequences:

$$x[n] = \{1, 2, 0, -4, -1\}, \quad h[n] = \{-2, 0, 1\}$$

length-N

length-M

$x[-k] = \{-1, -4, 0, 2, 1\}$

Start point = 0 + (-1)

x[n] start

h[n] start

= -1

$y = X h$

$h \in \mathbb{R}^3$ → length-3 vector

$y \in \mathbb{R}^7$ → length-(5 + 3 − 1) vector

N M

7

$X \in \mathbb{R}^{7 \times 3}$ → 7 × 3 matrix

$$y = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \\ -4 & 0 & 2 \\ -1 & -4 & 0 \\ 0 & -1 & -4 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 1 \\ 10 \\ 2 \\ -4 \\ -1 \end{bmatrix}$$

x[k]

h

Constant diagonals

n = -1

n = 0

$y[n] = \{-2, -4, 1, 10, 2, -4, -1\}$

Computing Convolution: Applying Convolution Sum

Suppose we would like to convolve the following two sequences: menti.com : 2105 8661

$$x[n] = \left(\frac{1}{2}\right)^n u[n],\ h[n] = \left(-\frac{3}{4}\right)^n u[n]$$

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$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k] \left(-\frac{3}{4}\right)^{n-k} u[n-k]$$



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$$\left(\frac{1}{2}\right)^k \left(-\frac{4}{3}\right)^k$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} u[k]\, u[n-k]$$

$$\left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} = \left(-\frac{3}{4}\right)^n \left(-\frac{4}{6}\right)^k$$

Computing Convolution: Applying Convolution Sum

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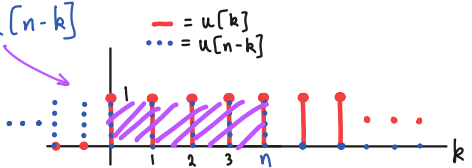
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Computing Convolution: Applying Convolution Sum

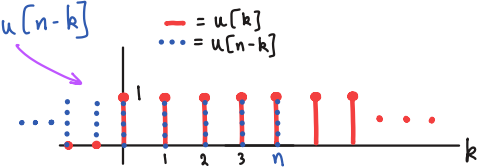
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$$x[n] = \left(\frac{1}{2}\right)^n u[n],\ h[n] = \left(-\frac{3}{4}\right)^n u[n]$$

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} u[k]\,u[n-k]$$

$$\left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} = \left(-\frac{3}{4}\right)^n \left(-\frac{4}{6}\right)^k$$

$$= \left(-\frac{3}{4}\right)^n \sum_{k=0}^{n} \left(-\frac{2}{3}\right)^k$$

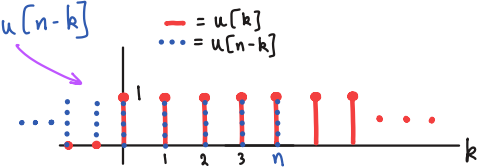


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$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} u[k]u[n-k]$$



$$\left(\frac{1}{2}\right)^k \left(-\frac{3}{4}\right)^{n-k} = \left(-\frac{3}{4}\right)^n \left(-\frac{4}{6}\right)^k$$

$$= \left(-\frac{3}{4}\right)^n \sum_{k=0}^{n} \left(-\frac{2}{3}\right)^k$$

$$\sum_{k=0}^{n} a^k = \frac{1-a^{n+1}}{1-a}$$

$$= \begin{cases} 0 & , n<0 \\ \left(-\frac{3}{4}\right)^n \left(\frac{1-\left(-\frac{2}{3}\right)^{n+1}}{1-\left(-\frac{2}{3}\right)}\right) & , n \geq 0 \end{cases}$$

Computing Convolution: Applying Convolution Sum

Suppose we would like to convolve the following two sequences:

$$x[n] = \left(\frac{1}{2}\right)^n u[n],\ h[n] = \left(-\frac{3}{4}\right)^n u[n]$$

$$x[n] * h[n] = \begin{cases} 0 & , n < 0 \\ \left(-\frac{3}{4}\right)^n \left(\frac{1-\left(-\frac{2}{3}\right)^{n+1}}{1-\left(-\frac{2}{3}\right)}\right) & , n \geq 0 \end{cases}$$

$$= \frac{3}{5}\left(-\frac{3}{4}\right)^n \left(1-\left(-\frac{2}{3}\right)^{n+1}\right) u[n]$$

Cases for Computing Convolution

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Three distinct cases for computing the convolution of two discrete-time signals:

1. Two finite-length signals
   - Shift and overlap or table method
   - Matrix method
2. Two infinite-length signals
   - Convolution
   - (z-transform)
3. One finite-length signal and one-infinite-length signal
   - LTI system properties and $\delta[n]$ properties

Computing Convolution: Finite and Infinite-length Sequences

Suppose we would like to convolve the following two sequences:

$$x[n] = \{-2, 0, \underset{\uparrow}{0}, 3, 0, 1\}, \; h[n] = e^{-n}u[n]$$

$h[n] * \delta[n] = h[n]$

$$y[n] = x[n] * h[n]$$

$$= (-2\delta[n+2] + 3\delta[n-1] + \delta[n-3]) * h[n]$$

$$= -2\delta[n+2] * h[n] + 3\delta[n-1] * h[n] + \delta[n-3] * h[n]$$

$$= -2h[n+2] + 3h[n-1] + h[n-3]$$

$$= -2e^{-(n+2)}u[n+2] + 3e^{-(n-1)}u[n-1] + e^{-(n-3)}u[n-3]$$

Extra Convolution Practice

Convolve the following two sequences:

$$x[n] = \{-3, 6, 2, \underset{\uparrow}{0}, 1, 1\}, \; h[n] = \{1, \underset{\uparrow}{-1}\}$$

$x[n]$ starts @ $n=-3$

$h[n]$ starts @ $n=-1$

↳ $y[n]$ starts @ $n=-4$

$y[n] = x[n] * h[n]$

$= Hx$

$x \in \mathbb{R}^6$

$y \in \mathbb{R}^7$

$H \in \mathbb{R}^{7\times 6}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ 6 \\ 2 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 9 \\ -4 \\ -2 \\ 1 \\ 0 \\ -1 \end{bmatrix} \begin{matrix} \leftarrow n=-4 \\ \\ \\ \\ \leftarrow n=0 \\ \\ \end{matrix}$$

H

$y[n] = \{-3, 9, -4, -2, \underset{\uparrow}{1}, 0, -1\}$

Extra Convolution Practice

Convolve the following two sequences:

$$x[n] = u[n],\ h[n] = \left(\frac{2}{3}\right)^{n-2} u[n-4]$$

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{2}{3}\right)^{k-2} u[k-4]\,u[n-k]$$

$$= \left(\frac{2}{3}\right)^{-2} \sum_{k=4}^{n} \left(\frac{2}{3}\right)^{k} \qquad n \geq 4$$

$n < 4$: $y[n] = 0$

Let $m = k-4$

$$y[n] = \left(\frac{2}{3}\right)^{-2} \sum_{m=0}^{n-4} \left(\frac{2}{3}\right)^{m+4}$$

$$= \left(\frac{2}{3}\right)^{-2} \left(\frac{2}{3}\right)^{4} \sum_{m=0}^{n-4} \left(\frac{2}{3}\right)^{m}$$

$$= \left(\frac{2}{3}\right)^{2} \cdot \left(\frac{1 - \left(\frac{2}{3}\right)^{n-3}}{1 - \left(\frac{2}{3}\right)} \right)$$

$$= \frac{4}{3}\left(1 - \left(\frac{2}{3}\right)^{n-3}\right) u[n-4]$$